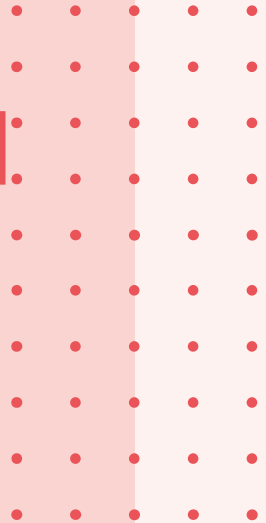
Variables & Data Types: Part 2!!!



Icebreaker Time!

2 Truths and a Lie

- Think of two fun facts about yourself and then another “fact” about you that isn’t true!
- Tell everyone else in your group your two facts and lie–it’s up to them to guess which one is your lie!
  - *Ex. I have a dog (truth), my favorite candy is Twizzlers (lie), my favorite type of music is pop (truth)*
- Have fun!



Women in Tech



Hedy Lamarr

Discuss...

What did Hedy Lamarr invent?

What did she do outside of technology?

What obstacles did she face as a woman?





Let's review what we did last week!

Data Types

integer

A non-decimal number
(Ex. 0, 123, -92)

float

A decimal number
(Ex. 3.14, 8.0, -0.17)

String

A word/phrase surrounded by quotations
Ex. (“hello”, “banana”, “puppy”)

bool

True or False

Variables

- Variables help us give names to different values
- Help us tell the difference between objects with the same data type!
  - *Ex. “banana” is different than “Banana”*
- Makes it easier to read code!
  - *Ex. favorite_thing = “To Code!”, this_year = 2024*
- Remember our rules for naming variables!
  - *Must start with a letter or an underscore*
  - *Variable names cannot have spaces–use an underscore (“_”) instead!*

Math/Logic Operators

- Addition: plus sign (+)
- Subtraction: minus sign (-)
- Multiplication: multiplication symbol (*)
- Division: division symbol (/)
- Modulo (%): tells us the remainder after dividing two numbers
    - *Ex. 10/3 = 3 r1, so 10 % 3 = 1*

Boolean/Comparison Operators

- *Equal to (==)*
  - *IMPORTANT: notice how we use 2 "=" signs–we only use one when we are creating/setting a variable!*
  - *Ex. love_cats == True*
- *NOT:*
  - *this negates (gives us the opposite of) a boolean*
  - *Ex. not True == False, not False == True*
- *Greater than (>)*
  - *Ex. 6 > 2*
- *Less than (<)*
  - *Ex. 5 < 10*

What would happen if we ran these three lines?

```
x = 3 + (5 * 2)  //don't forget parentheses first!

y = 6 % 5  //what is the remainder of 6/5?

print(x + y)
```

What would happen if we ran these three lines?

`x = 3 + (5 * 2)` *x stores 13*

`y = 6 % 5` *y stores 1*

`print(x + y)` *14 is printed (13 + 1 = 14)*

Conditional Statements

- AND: For this to be True, BOTH statements need to be True

  Is this True or False?

  22 == 22 and 43 > 44

  5 < 7 and 7 < 9

Conditional Statements

- **AND:** For this to be True, BOTH statements need to be True

  Is this True or False?

  22 == 22 and 43 > 44

  False: 22 equals 22, but 43 is less than 44. Both statements need to be true when using “and”

  5 < 7 and 7 < 9

  True: both statements are true

Conditional Statements

- OR: For this to be True, only one statement needs to be True

  Is this True or False?

  37 < 5 or 18 < 37

  77 == 7 or 6 < 3

Conditional Statements

- **OR:** For this to be True, only one statement needs to be True

  Is this True or False?

  37 < 5 or 18 < 37

  True: 37 is not less than 5, but 18 is less than 37. At least statement needs to be True when using “or”

  77 == 7 or 6 < 3

  False: both statements are false

Quick Break!



Practice Exercises

https://tinyurl.com/59a75bx7



Thank You!

(see you next week!)